

# Ankit Ashok

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## EDUCATION

### Birla Institute of Technology and Science (BITS), Pilani

Dual Major in B.E. Mechanical Engineering and M.Sc. Mathematics

Hyderabad, India

2019 - 2024

Cumulative GPA: 7.8/10

Relevant Coursework: Linear Algebra (MATH II), Statistical Inference and Applications, Numerical Analysis, Optimization, Applied Statistical Methods, Artificial Intelligence for Robotics, Data Structures and Algorithms.

## WORK EXPERIENCE

### AI Engineer, Deloitte Consulting

Hyderabad, India

Aug 2024 - present

- Owned and deployed **LangGraph-based agentic workflows on AWS AgentCore** for insurance and medical document processing, integrating **multimodal OCR, MCP servers and LLMs** to automate decisioning at scale, reducing operational workload by 90%.
- Architected and delivered a production-grade AWS **data ingestion** platform (S3, SQS, Lambda, Glue, RDS) with CDC-driven pipelines, enabling real-time, compliant **claims processing** and reducing manual data preparation by 80%.
- Led development of LangGraph **agents for dynamic claims adjudication** by embedding **regulatory logic and SME-driven rules**, driving scalable and compliant decision-making across workflows.
- Scaled and enhanced a GenAI-powered business glossary tool with RAG-based context retrieval, data quality generation, and anomaly detection, **improving data-driven decision-making for 20+ enterprise clients**.

### Data Science Intern, Genpact

Bengaluru, India

Feb 2024 - Jul 2024

- Collaborated with client-side managers of a prominent media company to adapt and validate machine learning pipelines using Python and R for **2 new regions**, enhancing predictive accuracy and business insights.
- Optimized market mix modelling workflows by automating post-processing tasks in Python, **reducing manual intervention by 70%**, saving the team approximately **10 hours per week**.

### Quant Intern, Futures First

Gurgaon, India

Jul 2023 - Dec 2023

- Introduced an **LSTM-based model** for pre-macroeconomic data trading, delivering a **30-second execution edge with 70% directional accuracy**.
- Developed time series (SARIMA, ARIMAX) and calibrated stochastic interest rate models (Hull-White, CIR) using MLE & Monte Carlo, **reducing RMSE by 80%** across various interest rate scenarios.

### Research Intern @ TavLab, Indraprastha Institute of Information Technology

New Delhi, India

Jun 2022 - Oct 2022

- Analysed the entropy data of **BERT** and w2v embeddings of spike protein sequences of COVID-19 using line plots, boxplots and detrended cross-correlation analysis in Python, R and Excel and predicted case surges using Random Forest with an **R2 value of 0.75**.

## PROJECTS

### Prediction of High Temperature Deformation Behaviour of $\beta$ -Titanium Alloy using Various Machine Learning Approaches [\[Repo Link\]](#)

Jan 2023 - Jul 2023

- Implemented a Machine Learning workflow involving data cleaning, hyperparameter optimization and result validation in Python to study the flow stress of  $\beta$ -Titanium Alloy and found the most effective model, XGBoost with a **MAPE of 3.18%** and compared its performance to a physical mathematical model.

### Catering Robot Simulation [\[Repo Link\]](#)

Sept 2022 - Dec 2022

- Optimised the spacing of a restaurant-like environment for the motion of a robot by implementing FastSLAM and D\* Lite algorithms in Python and presented it graphically along with a simulation of the robot's motion using a box2d and matplotlib based physics environment.

## TECHNICAL SKILLS

- Programming Languages: Python, SQL, R
- Libraries and Frameworks: NumPy, pandas, Matplotlib, scikit-learn, TensorFlow, PyTorch, LangChain, LlamaIndex, OpenAI, Anthropic, LangGraph, Pydantic AI, AWS AgentCore, MCP
- Big Data, Querying and VectorDB: PySpark, PostgreSQL, MySQL, Milvus, Pinecone
- Cloud & MLOps: Azure - (ML, Functions, Blob Storage, DevOps), AWS - (Sagemaker, Bedrock, Lambda, Glue, S3, EC2, SQS, DynamoDB)
- Tools & Version Control: Git, Docker, Microsoft Excel, PowerBI

## ADDITIONAL

- Certifications:** [Microsoft Certified: \(Azure Data Scientist Associate, Azure Fundamentals\)](#), [AWS Certified AI Practitioner](#).
- Publication:** Ashok, A. et al. (2023) 'Strength and Fracture Energy Dependence of Additively Manufactured Polymer Parts on Build Orientation, Density, and Layer Thickness', in IMECE2023. Volume 4: Advanced Materials: Design, Processing, Characterization and Applications; Advances in Aerospace Technology. Available at: <https://doi.org/10.1115/IMECE2023-112241>.
- Awards & Recognition:** Outstanding Performance Award (Deloitte, October 2025), Applause Award (Deloitte, April 2026)